MC 501 Extended Codebook and Reliability Protocol: Worked Example

- Rebuilt from my own research project, Play to Learn in VR (P01.01), continuing the student study from the Research Proposal and Definitions Practice examples: one coder reading 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit: the codebook and the reliability protocol as one document with two headed parts. Every example tweet is paraphrased from a real tweet in the archive; none is quoted, so a search cannot lead back to the account.
- The rubric map at the end shows which grading row each part earns and why.

The example

Part 1: Codebook

Study overview

This codebook measures whether tweets about playing in virtual reality (VR) that refer to learning are more often positive toward VR than tweets that do not (H1), which benefits and obstacles of VR the learning tweets name (RQ1), and whether the balance of benefits to obstacles differs between learning and other tweets, the comparison that separates affordance theory from the cognitive-load account (Mayer et al., 2023) (RQ2). The material is English-language tweets, not retweets, that mention VR together with play, drawn from the Play to Learn in VR archive collected through the Twitter Academic Research API. The time period is calendar year 2021, January 1 through December 31.

Unit of analysis

One tweet: the text the author wrote in that single post. A quoted tweet, a linked page, an image, and the replies underneath are not part of the unit. Every variable below is coded on one tweet. Each tweet is also recorded with three fields that are looked up, not coded: its tweet ID, its posting date, and its like count. The RQ2 balance (benefit only, obstacle only, both, or neither) is computed in R from variables 3 and 4 for the same tweet, so it adds no coding decision.

Variable 1: Learning reference

Element	Learning reference

Element	Learning reference
Conceptual definition and source	Whether the tweet is about someone gaining or passing on a skill or knowledge in or through VR: learning, teaching, studying, practicing, or being trained. This definition is my own. The published discourse studies defined their learning tweets by a word list (Lampropoulos et al., 2022), and I define the variable by what the tweet describes instead, because in this archive the same words appear in game titles, adult-game advertisements, and idioms, and a word list cannot tell those apart from a person learning.
Operational definition	Read the whole tweet. Mark 1 if any clause describes a person (the writer, a named person, learners in general, or a customer in an advertisement) learning, teaching, practicing, studying, or being trained in something, and the activity happens in or through VR. Otherwise mark 0. If the tweet contains a learning word, check what the word means in its sentence before marking.
Level of measurement, and what it allows	Nominal: two named categories with no order. It allows counts and percentages and a chi-square test against another nominal or ordinal variable. It rules out a mean, because 0 and 1 are labels and not amounts.
Coded values	0 = no learning reference. 1 = learning reference.
Edge-case decision rules	1. A learning word that is part of a game or product title (a shooting game called a trainer, an app called a trainer) or part of an idiom (hype train, learned the hard way, learned my lesson) is coded 0 unless another clause describes learning. 2. "Learn" meaning "find out about" (learn which headset games were played most this year) is coded 0. 3. A tweet that only shares a video title is coded 1 when the title describes someone teaching or learning in VR (a video in which one avatar teaches another how to do something in a social VR app), because the unit is the tweet text and the text describes learning. 4. An advertisement for a product that teaches (a VR app that teaches you to play the piano) is coded 1. The variable records that learning is described, not who describes it or why. 5. Machine-learning and artificial-intelligence hashtags are coded 0: the learning is a computer's, not a person's. 6. Learning to play a VR game, or relearning it on new controllers, is coded 1, because the skill is gained in VR.
Validity sentence	I would doubt this variable if most of the tweets coded 1 turned out to be companies advertising learning apps, because it would then be measuring marketing copy rather than people describing learning.

Variable 2: Tone toward VR

Element	Tone toward VR
Conceptual definition and source	The writer's evaluation of VR or of the VR experience the tweet describes: whether the tweet speaks well of it, badly of it, or neither. This is sentiment in the sense the discourse studies use, the positive or negative evaluation carried by the words of a text (Lampropoulos et al., 2022; Mohammad & Turney, 2013), read here by a person rather than counted by a lexicon.
Operational definition	Find every word, phrase, or emoji in the tweet that evaluates VR, a VR headset, or a VR experience, including a game played in VR (fun, love, amazing; nauseous, broken, waste of money). Mark 3 when every evaluation is favorable, 1 when every evaluation is unfavorable, and 2 when the tweet has both or has none. Evaluations of something other than VR (a friend, a game that is not in VR, a public figure) do not count.
Level of measurement, and what it allows	Ordinal: three categories in order from negative to positive, with no claim that the steps are equal. It allows ordered comparisons (a larger share of learning tweets sits at positive), a median and a mode, and a chi-square test. It rules out a mean, because the distance from negative to neutral is not known to equal the distance from neutral to positive.
Coded values	1 = negative. 2 = neutral or mixed. 3 = positive.
Edge-case decision rules	1. Praise for a game played in VR counts as an evaluation of a VR experience. Praise for a game played outside VR, a streamer, or a friend does not count, so a tweet with only that praise is coded 2. 2. An advertisement that lists features or a release date with no evaluative word is coded 2. An advertisement that calls the experience amazing is coded 3, because the variable records what the tweet says, not whether the writer is sincere. 3. A complaint the writer laughs off (a laughing emoji after describing a glitch that threw them out of a match) is coded by its literal words, so it is 1 unless a favorable clause also appears, which makes it 2. 4. Emoji count as evaluations: a heart or a fire emoji attached to VR is favorable, and a vomiting face is unfavorable.
Validity sentence	I would doubt this variable if the positive codes were mostly promotional tweets, because advertising language is favorable by design and would say nothing about how users feel about VR.

Variable 3: Affordance named

Element	Affordance named

Element	Affordance named
Conceptual definition and source	The benefit the tweet attributes to VR: something VR lets a person do, or do better, in the writer's account. In affordance theory an affordance is what an environment offers a person to do (Gibson, 1979), and the categories below follow the benefits users and students name most often, engagement, interaction, collaboration, and skill practice (Allcoat & von Mühlenen, 2018; Laine & Lee, 2024).
Operational definition	Look for a clause that says what VR, a headset, or a VR experience lets someone do or makes better. Match that clause to one category below. If the tweet names benefits in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a benefit, mark 0.
Level of measurement, and what it allows	Nominal: named categories with no order. It allows counts, percentages, and a chi-square test. It rules out a mean and a median, because the categories cannot be ranked.
Coded values	0 = no benefit named. 1 = fun or engagement (enjoyment, excitement, being absorbed). 2 = hands-on interaction (moving, building, handling objects, being inside the scene). 3 = doing it with others (playing, meeting, or learning alongside other people). 4 = practicing a skill (getting better at something, in a game or outside it, in a safe setting). 5 = other benefit.
Edge-case decision rules	1. A feature named without saying what it lets a person do (multiplayer, hand tracking, a high refresh rate) is coded 0. A feature is not a benefit until the tweet says what it is for. 2. "Fun" or "love it" with no more detail is coded 1. 3. Learning to play a VR game better is coded 4, the same as practicing piano in VR, because the variable records the kind of benefit, not whether it carries outside the game. 4. A tweet that praises VR for company is coded 3 only when other people are involved; a solo experience described as absorbing is coded 1.
Validity sentence	I would doubt this variable if more than a third of the tweets that name a benefit landed in "other benefit," because the categories would then not be describing what users actually name.

Variable 4: Constraint named

Element	Constraint named

Element	Constraint named
Conceptual definition and source	The obstacle the tweet attributes to VR: something that stopped, limited, or discouraged the writer or someone else from doing something in VR. A constraint is this study's counterpart to an affordance (Gibson, 1979), what the environment blocks, and the categories follow the barriers the reviews report most often, cybersickness, accessibility, and hardware (Sunardi et al., 2025; van der Meer et al., 2023).
Operational definition	Look for a clause that names a problem with VR, the hardware, or the experience that limited or discouraged use. Match it to one category below. If the tweet names problems in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a problem, mark 0.
Level of measurement, and what it allows	Nominal: named categories with no order. It allows counts, percentages, and a chi-square test. It rules out a mean and a median.
Coded values	0 = no obstacle named. 1 = physical discomfort (motion sickness, nausea, eye strain, headache, fatigue). 2 = hardware cost (the price of a headset or computer, cannot afford one). 3 = technical performance (lag, crashes, glitches, setup trouble, incompatibility). 4 = other obstacle (space, hygiene, other players' behavior, anything else).
Edge-case decision rules	1. A problem with something other than VR (a slow phone, a friend's lost username) is coded 0 unless the tweet ties the problem to VR. 2. A warning or advice about a problem (how to build up tolerance to motion sickness) is coded in that category even when the writer did not suffer it, because the tweet names it as an obstacle. 3. Being thrown out of a match by a glitch is coded 3; what the writer lost as a result does not change the category. A problem the writer only jokes about expecting someday, without saying it has limited anyone yet, is coded 0. 4. "I can't play, no headset yet" is coded 2 only when price or affordability is stated; otherwise it is coded 4.
Validity sentence	I would doubt this variable if I only ever coded an obstacle when the tweet's tone was negative, because the variable would then be copying tone rather than recording which obstacle was named.

Decision rules for the whole codebook

These rules settle the cases that did not sort themselves while I read 30 immersion tweets drawn at random from the archive and kept out of the study sample. Each one applies before or across the four variables.

1. Before coding, check that the tweet is about virtual reality. If VR appears only inside another word, a username, or an unrelated abbreviation, and the tweet is about something else (a sports position, a rhythm game played on a monitor), mark it out of frame, code nothing, and replace it with the next tweet on the reserve list.

2. If two tweets in the sample carry the same text word for word (a bot reposting one promotion), code the first one and mark the second a duplicate; it is replaced from the reserve list.

3. The archive strips digits and some punctuation, so a console name or a date can appear with its number missing. Code the words that remain and never fill in a missing word from a guess.

4. A tweet that is only a string of hashtags, with no sentence, is coded 0 on learning reference, 2 on tone, 0 on affordance, and 0 on constraint.

5. Code what the tweet says, not who sent it. An advertisement, a non-fungible token (NFT) promotion, or an adult-game listing that slipped past the archive's sensitive flag is coded by its words like any other tweet. A tweet whose text is sexually explicit is marked out of frame and replaced, and it is never used as an example.

6. When one clause fits two variables, code it on each. A sentence saying friendly strangers taught the writer to play is a learning reference (1) and an affordance of doing it with others (3).

7. Code the username, the follower count, and the like count hidden. The like count is looked up only after all four variables are coded for that tweet.

Examples

Category	Two or three examples, each paraphrased from a tweet in the archive
Learning reference 1	(a) A new headset owner says friendly strangers in a VR tag game taught them how to play.
	(b) A parent says a multiplayer VR puzzle game lets them and their son play and learn logic together.
	(c) A university announcement says new VR training facilities let students practice clinical skills in a safe setting.
Learning reference 0	(a) A promotion invites readers to learn more about an upcoming VR shooting demo.
	(b) A string of coding and technology hashtags that puts machine learning beside VR.
	(c) A film synopsis about a game designer and a marketing trainee trapped inside a VR game.

Category	Two or three examples, each paraphrased from a tweet in the archive
Tone 3, positive	(a) A player calls a VR horror game great and tells a friend to play it.
	(b) A player calls a VR demo amazing from start to finish and shares a recorded fight.
	(c) A gamer calls a VR title one of the most immersive they have tried.
Tone 2, neutral or mixed	(a) A student asks whether a cabled headset will reach the hallway, since their study room has no space (no evaluation).
	(b) A recap of a console maker's announcement stream lists upcoming games, some in VR, with no evaluative word (no evaluation).
	(c) A player says every VR owner must try one game, but the game needs better filters for eye strain (both).
Tone 1, negative	(a) A player says the motion controllers always bug out, which is frustrating, so they rarely use their VR.
	(b) A writer says they do not like the idea of working and meeting in VR and worries the next generation will stop socializing.
	(c) A buyer says motion sickness will take time to get used to and the headset costs too much for what it plays.
Affordance 0, none	(a) A player says they are about to buy the same horror game yet again to play it in VR.
	(b) A recap of a console maker's announcement stream lists upcoming games, some in VR.
Affordance 1, fun or engagement	(a) A player calls a VR horror game great and spooky.
	(b) A gamer calls a VR title one of the most immersive they have tried.
	(c) A player who replaced a broken headset says they are excited to be slicing blocks in a rhythm game again.
Affordance 2, hands-on interaction	(a) A writer wonders whether a VR driving game gives the feel of the road and of acceleration, the control sensations people play for.
	(b) A social VR user says the point of VR for them is that it gets them standing up instead of sitting for hours.
Affordance 3, doing it with others	(a) A player says they want to spend more time in a social VR app with friends.
	(b) A group that cannot meet in person plans to play a tabletop game together in VR.
	(c) A player describes a day of multiplayer VR with children who coached them through a squad game.

Category	Two or three examples, each paraphrased from a tweet in the archive
Affordance 4, practicing a skill	(a) A VR app promises students practice at interviews, presentations, and public speaking.
	(b) A university announcement says VR training lets students practice clinical skills in a safe setting.
	(c) A player advises a newcomer to practice a VR game in multiplayer first to learn it.
Affordance 5, other benefit	(a) A player tells themselves an all-day rhythm game session counts as exercise.
	(b) A parent says a VR workout app makes them feel powerful again.
	(c) A player says VR has been a big help to them through lockdown.
Constraint 0, none	(a) A player calls a VR horror game great and tells a friend to play it.
	(b) A player calls a VR demo amazing from start to finish.
Constraint 1, physical discomfort	(a) A new headset owner says they are lying down after an hour of a VR tag game because they feel motion sick.
	(b) A player says a cockpit with a lot of roll and pitch is one of the few things that makes them nauseous in VR.
	(c) A player advises newcomers to let themselves get sick, rest, and go back in until the body adjusts.
Constraint 2, hardware cost	(a) A player says they play a game without a headset because they cannot afford one.
	(b) A social VR user wants a better headset someday but says they are expensive.
Constraint 3, technical performance	(a) A player says the motion controllers always bug out, so they rarely use their VR.
	(b) A player says their boundary setup resets or puts the floor at the wrong height almost every time they start a VR game.
Constraint 4, other obstacle	(a) A student asks whether a cabled headset will reach the hallway, since their study room has no space.
	(b) A parent says their child skipped a shared VR headset because of germs from everyone wearing it.

Part 2: Reliability protocol

I am the only coder. Every reliability claim in this protocol is a claim about my own consistency over time, intracoder reliability (stability), and not agreement between two coders.

6. The statistic

I will report Krippendorff's alpha for each variable, with the nominal metric for learning reference, affordance, and constraint and the ordinal metric for tone. I chose it over Cohen's kappa because one statistic then covers all four variables and respects tone's order, so a shift from positive to neutral counts as a smaller disagreement than a shift from positive to negative. This is intracoder reliability (stability): the two passes are mine, so alpha here measures how consistently I apply the codebook, not whether two coders agree. I will also report percent agreement, but not on its own, because it ignores the agreement that chance alone produces (Lombard et al., 2002). That matters here: most tweets name no benefit and no obstacle, so two passes could agree on 0 most of the time while disagreeing on nearly every tweet that does name one.

7. The threshold

A variable passes at alpha of .80 or higher, the level Lombard et al. (2002) call acceptable in most situations. A stability figure is a weaker claim than agreement between two coders, because it cannot catch a reading I make the same way both times, so I will report it as stability and never as if it were intercoder agreement.

8. The stability procedure

I will re-code 15 tweets, 10 percent of the 150, at least 48 hours after their first coding. Before the second pass, I copy those 15 tweet IDs and texts into a new coding form with no code columns, so my first codes stay hidden until both passes are done. I then compute alpha and percent agreement in R from the two forms joined by tweet ID.

9. A fresh reliability sample

The 15 come from tweets that played no part in building or revising the codebook. The codebook was built on 30 immersion tweets, and the pilot of 25 tweets in the Sampling Plan will be used to revise it, so both sets are excluded. After I freeze the codebook, I draw the 15 at random in R, with the seed written in the script, from the 125 sample tweets that are not in the pilot: 8 from the learning-word half and 7 from the other half, so both groups H1 compares are represented.

10. When it fails

If a variable falls below .80, I revise that variable only. I read every disagreement for it in my disagreement log, a sheet with the tweet ID, both codes, and the rule each pass applied, and I rewrite or add only the rules where disagreements cluster, without changing any category's conceptual definition. Two guards keep the codebook from bending toward my habits. First, the revised rules are written and dated before I look at any new tweet. Second, I test them on a new fresh sample of 15 drawn the same way from tweets not yet used for reliability, never on the 15 that failed, and I re-code all 150 tweets on that variable under

the revised rules. I allow two revision rounds. A variable still below .80 after the second round is reported as unreliable and is not used in any test.

11. Reporting a disappointing number

I will report every figure I compute, for every variable and every round, including any that miss .80. For a low figure I will say that my own reading of that variable shifted across two days, name which result rests on it (for tone, that is H1), and describe that result as tentative.

12. Single-coder safeguards

1. Rules before coding: I wrote every category's inclusion and exclusion rule in Part 1 before coding the study sample.

2. Examples from my material: each category carries two or three examples paraphrased from real tweets in the archive, and I check a hard case against them before coding it.

3. A frozen codebook with a change log: I freeze the codebook after the pilot and log every later change with its date, the rule changed, and the reason.

4. An ambiguous-case memo: each tweet I cannot place at once goes into a memo with the rule I applied, and I settle it against the written rule, not by feel.

5. A fixed order with the source hidden: I code in tweet-ID order with the username, follower count, and like count hidden until all four variables are coded.

13. Second-coder counterfactual

With a second coder, I would double-code 30 tweets, 20 percent of the sample, drawn the same way as the fresh reliability sample, and report Krippendorff's alpha for each variable as intercoder agreement. I would train the coder on this codebook with 20 practice tweets from outside the sample, meet to compare codes, and settle each disagreement against the written rule, logging it and revising the rule if the rule caused it. The finding most at risk is H1, because it rests on tone, the variable that needs the most judgment, and the line between neutral and mixed is where two coders would most often part; RQ2 is next, because it depends on whether a clause counts as a benefit at all.

References

- Allcoat, D., & von Mühlenen, A. (2018). Learning in virtual reality: Effects on performance, emotion and engagement. *Research in Learning Technology, 26*. https://doi.org/10.25304/rlt.v26.2140
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.
- Laine, T. H., & Lee, W. (2024). Collaborative virtual reality in higher education: Students' perceptions on presence, challenges, affordances, and potential. *IEEE*

Transactions on Learning Technologies, 17, 280-293. https://doi.org/10.1109/TLT.2023.3319628

- Lampropoulos, G., Keramopoulos, E., Diamantaras, K., & Evangelidis, G. (2022). Augmented reality and virtual reality in education: Public perspectives, sentiments, attitudes, and discourses. *Education Sciences, 12*(11), Article 798. https://doi.org/10.3390/educsci12110798
- Lombard, M., Snyder-Duch, J., & Bracken, C. C. (2002). Content analysis in mass communication: Assessment and reporting of intercoder reliability. *Human Communication Research, 28*(4), 587-604. https://doi.org/10.1111/j.1468-2958.2002.tb00826.x
- Mayer, R. E., Makransky, G., & Parong, J. (2023). The promise and pitfalls of learning in immersive virtual reality. *International Journal of Human-Computer Interaction, 39*(11), 2229-2238. https://doi.org/10.1080/10447318.2022.2108563
- Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. *Computational Intelligence, 29*(3), 436-465. https://doi.org/10.1111/j.1467-8640.2012.00460.x
- Sunardi, Meyliana, Hendric, S. W. H. L., & Rosmansyah, Y. (2025). Exploring immersive virtual reality in higher education: Research gap and future direction, a scoping review. *IEEE Access, 13*, 76308-76321. https://doi.org/10.1109/ACCESS.2025.3565385
- van der Meer, N., van der Werf, V., Brinkman, W.-P., & Specht, M. (2023). Virtual reality and collaborative learning: A systematic literature review. *Frontiers in Virtual Reality, 4*, Article 1159905. https://doi.org/10.3389/frvir.2023.1159905

Rubric map

Rubric row	Points	Where the example earns it	Why it earns the full row
Codebook: study overview	4	The study overview paragraph	It names the question (H1, RQ1, and RQ2, with the rival accounts RQ2 separates), the material (English tweets about playing in VR from a named archive and collection route), and the time period (calendar 2021).
Codebook: unit of analysis	10	The unit of analysis paragraph	One coded item is stated exactly, once: one tweet's own text, with what is not part of it named. Every variable, the looked-up fields, and the computed RQ2 balance describe that same tweet.

Rubric row	Points	Where the example earns it	Why it earns the full row
Codebook: every variable	10	The four variable tables	Every variable carries all six elements from Definitions Practice: conceptual definition with its source, operational definition, level of measurement and what it allows, coded values, edge-case rules, and a validity sentence specific to that variable.
Codebook: decision rules	8	The seven decision rules	Each is written as a rule a coder can apply, and each covers a hard call met in the material: off-topic matches, a reposted promotion, stripped digits, hashtag strings, advertisements, one clause that fits two variables, and coding blind to the source.
Codebook: examples	8	The examples table	Every category of all four variables has two or three examples, each paraphrased from a real tweet in the archive, none invented.
6. The statistic	6	Section 6	It names Krippendorff's alpha with the reason for it over Cohen's kappa, calls it intracoder reliability (stability) in those words, and says why percent agreement alone is not enough, with the reason that bites in this data: most tweets are coded 0 on two variables.
7. The threshold	3	Section 7	A threshold (.80) with its source, and the sentence saying a stability figure is a weaker claim than two-coder agreement and is never reported as one.
8. The stability procedure	3	Section 8	All three conditions: 15 tweets, which is 10 percent of the 150; at least 48 hours later; first codes hidden by a fresh form with no code columns.
9. A fresh reliability sample	5	Section 9	The sample is kept apart from both sets used to build and revise the codebook (the immersion tweets and the pilot), and its size and the way it is drawn (seeded, from the 125 non-pilot tweets, split across the two groups) are stated.

Rubric row	Points	Where the example earns it	Why it earns the full row
10. When it fails	5	Section 10	It says in advance which categories are revised (only the failing variable's, where disagreements cluster), how (from the disagreement log, rules only), and two guards against fitting the codebook to habits: rules dated before new tweets are seen, and a new fresh sample for the retest.
11. Reporting a disappointing number	3	Section 11	It commits to reporting every figure, and says what it will say about a low one: that the reading shifted, which result rests on it, and that the result is tentative.
12. Single-coder safeguards	5	Section 12	Five safeguards, above the minimum of three, each named and described as something done in this study.
13. Second-coder counterfactual	5	Section 13	One paragraph with all four parts: 30 tweets double-coded, Krippendorff's alpha reported, training on 20 practice tweets with disagreements settled against the rule, and H1 named as the finding most at risk, with the reason.

The codebook rows total 40 and the protocol rows 35, graded together as one 75-point assignment.

The codebook revises the Definitions Practice entries where the 30 immersion tweets showed a gap: tone now counts a game played in VR as a VR experience, learning reference gained the rule for relearning a game, and constraint gained the rule for problems the writer only expects.